

* Squares - 1-150 (exam)
1-30 (Ratta)

Steps:

- ① (Last digit)²
- ② (Product) x 2
- ③ (First Group)²

eg- 24²

$$\begin{array}{r} 4 \overline{) 16 / 16} \\ \underline{576} \end{array}$$

Eg-1) 63²

3969

2) 142²

20164

3) 32²

1024

4) 43²

1849

5) 54²

2916

6) 61²

3721

7) 72²

5184

8) 83²

6889

9) 94²

8836

10) 101²

10201

11) 112²

12544

12) 123²

15129

13) 134²

~~1756~~
18356

14) 33²

1089

15) 44²

~~1936~~
1936

16) 51²

2601

17) 62²

3844

18) 73²

5329

19) 84²

7056

20) 91²

8281

21) 102²

10404

22) 113²

12769

23) 124²

15376

24) 131²

17161

25) 34²

1156

26) 41²

1761

27) 52²

2704

28) 63²

3969

29) 74²

5476

30) 81²

6561

31) 92²

8464

32) 103²

10609

33) 114²

12996

34) 121²

14641

35) 132²

17424

$$\begin{array}{r} 36^2 \\ 1296 \end{array}$$

$$\begin{array}{r} 47^2 \\ 2209 \end{array}$$

$$\begin{array}{r} 58^2 \\ 3364 \end{array}$$

$$\begin{array}{r} 69^2 \\ 4761 \end{array}$$

$$\begin{array}{r} 76^2 \\ 5776 \end{array}$$

$$\begin{array}{r} 87^2 \\ 7569 \end{array}$$

$$\begin{array}{r} 98^2 \\ 9604 \end{array}$$

$$\begin{array}{r} 109^2 \\ 11881 \end{array}$$

$$\begin{array}{r} 116^2 \\ 13456 \end{array}$$

$$\begin{array}{r} 127^2 \\ 16129 \end{array}$$

$$\begin{array}{r} 138^2 \\ 19044 \end{array}$$

$$\begin{array}{r} 149^2 \\ 21201 \end{array}$$

$$\begin{array}{r} 37^2 \\ 1369 \end{array}$$

$$\begin{array}{r} 48^2 \\ 2304 \end{array}$$

$$\begin{array}{r} 59^2 \\ 3481 \end{array}$$

$$\begin{array}{r} 66^2 \\ 4356 \end{array}$$

$$\begin{array}{r} 77^2 \\ 5929 \end{array}$$

$$\begin{array}{r} 88^2 \\ 7744 \end{array}$$

$$\begin{array}{r} 99^2 \\ 9801 \end{array}$$

$$\begin{array}{r} 106^2 \\ 11236 \end{array}$$

$$\begin{array}{r} 117^2 \\ 13689 \end{array}$$

$$\begin{array}{r} 128^2 \\ 16384 \end{array}$$

$$\begin{array}{r} 139^2 \\ 19221 \end{array}$$

$$\begin{array}{r} 146^2 \\ 21316 \end{array}$$

$$\begin{array}{r} 31^2 \\ 961 \end{array}$$

$$\begin{array}{r} 49^2 \\ 2401 \end{array}$$

$$\begin{array}{r} 56^2 \\ 3136 \end{array}$$

$$\begin{array}{r} 67^2 \\ 4489 \end{array}$$

$$\begin{array}{r} 78^2 \\ 6084 \end{array}$$

$$\begin{array}{r} 89^2 \\ 7921 \end{array}$$

$$\begin{array}{r} 96^2 \\ 9216 \end{array}$$

$$\begin{array}{r} 107^2 \\ 11449 \end{array}$$

$$\begin{array}{r} 118^2 \\ 13924 \end{array}$$

$$\begin{array}{r} 129^2 \\ 16641 \end{array}$$

$$\begin{array}{r} 136^2 \\ 18496 \end{array}$$

$$\begin{array}{r} 147^2 \\ 21609 \end{array}$$

* Unit digit '5'

$(65)^2$ 35^2 95^2 75^2
 4225 1225 9025 5625

45^2 65^2 85^2 55^2
 2025 4225 7225 3025

135^2 185^2 115^2 145^2
 18225 34225 13225 21025

Tables:-

HW

① 14, 15

② Squares (1 - 30)

14	14	15	15	14	14	15	15	14	15
28	28	30	30	28	28	30	30	28	30
42	42	45	45	42	42	45	45	42	45
56	56	60	60	56	56	60	60	56	60
70	70	75	75	84	70	75	75	70	75
84	84	90	90	98	84	90	90	84	90
98	98	105	105	112	98	105	105	98	105
112	112	120	120	126	112	120	120	112	120
126	126	135	135	140	126	135	135	126	135
140	140	150	150		140	150	150	140	150

1 → 1	11 → 121	21 → 441	1 → 1	11 → 121	21 → 441
2 → 4	12 → 144	22 → 484	2 → 4	12 → 144	22 → 484
3 → 9	13 → 169	23 → 529	3 → 9	13 → 169	23 → 529
4 → 16	14 → 196	24 → 576	4 → 16	14 → 196	24 → 576
5 → 25	15 → 225	25 → 625	5 → 25	15 → 225	25 → 625
6 → 36	16 → 256	26 → 676	6 → 36	16 → 256	26 → 676
7 → 49	17 → 289	27 → 729	7 → 49	17 → 289	27 → 729
8 → 64	18 → 324	28 → 784	8 → 64	18 → 324	28 → 784
9 → 81	19 → 361	29 → 841	9 → 81	19 → 361	29 → 841
10 → 100	20 → 400	30 → 900	10 → 100	20 → 400	30 → 900

③ Find squares of given nos

$$\begin{array}{r} 36 \\ 1296 \end{array}$$

$$\begin{array}{r} 168 \\ 28224 \end{array}$$

$$\begin{array}{r} 49 \\ 2401 \end{array}$$

$$\begin{array}{r} 102 \\ 10404 \end{array}$$

$$\begin{array}{r} 75 \\ 5625 \end{array}$$

$$\begin{array}{r} 87 \\ 7569 \end{array}$$

$$\begin{array}{r} 157 \\ 24649 \end{array}$$

$$\begin{array}{r} 86 \\ 7396 \end{array}$$

$$\begin{array}{r} 81 \\ 6561 \end{array}$$

$$\begin{array}{r} 93 \\ 8649 \end{array}$$

$$\begin{array}{r} 77 \\ 5929 \end{array}$$

$$\begin{array}{r} 80 \\ 6400 \end{array}$$

$$\begin{array}{r} 66 \\ 4356 \end{array}$$

$$\begin{array}{r} 124 \\ 15376 \end{array}$$

$$\begin{array}{r} 121 \\ 14641 \end{array}$$

$$\begin{array}{r} 38 \\ 1444 \end{array}$$

$$\begin{array}{r} 101 \\ 10201 \end{array}$$

$$\begin{array}{r} 78 \\ 6084 \end{array}$$

$$\begin{array}{r} 97 \\ 9409 \end{array}$$

$$\begin{array}{r} 139 \\ 19321 \end{array}$$

$$\begin{array}{r} 175 \\ 30625 \end{array}$$

$$\begin{array}{r} 113 \\ 12769 \end{array}$$

$$\begin{array}{r} 142 \\ 20164 \end{array}$$

$$\begin{array}{r} 177 \\ 31329 \end{array}$$

$$\begin{array}{r} 119 \\ 14161 \end{array}$$

$$\begin{array}{r} 167 \\ 27889 \end{array}$$

$$\begin{array}{r} 186 \\ 34596 \end{array}$$

$$\begin{array}{r} 138 \\ 19044 \end{array}$$

$$\begin{array}{r} 112 \\ 12544 \end{array}$$

$$\begin{array}{r} 173 \\ 29929 \end{array}$$

$$\begin{array}{r} 77 \\ 5929 \end{array}$$

$$\begin{array}{r} 125 \\ 15625 \end{array}$$

$$\begin{array}{r} 161 \\ 25921 \end{array}$$

3844

686

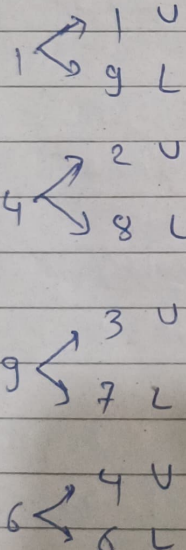
4225

2473665
4225103
10609189
33856117
13689156
~~23736~~
24336189
33856135
18225151
22801119
14161142
20164159
25281111
~~10541~~ 12321

Session - 3.

10/09/2026

* HW: Tables 1-16
Squares 1-30
Cubes 1-16

*


Unit digit	Possibility
1	1, 9
4	2, 6
9	3, 7
6	4, 6
5	5
0	0

* Upper Order

eg- 1) $\sqrt{3969} = 63$ 2) $\sqrt{14641} = 121$ 3) $\sqrt{17424} = 132$ 4) $\sqrt{7056} = 84$

83
87

2225

$$\begin{array}{r} \sqrt{5041} \\ 71 \end{array}$$

$$\begin{array}{r} \sqrt{10201} \\ 101 \end{array}$$

$$\begin{array}{r} \sqrt{1764} \\ 42 \end{array}$$

$$\begin{array}{r} \sqrt{8464} \\ 92 \end{array}$$

$$\begin{array}{r} \sqrt{2809} \\ 53 \end{array}$$

$$\begin{array}{r} \sqrt{17689} \\ 133 \end{array}$$

$$\begin{array}{r} \sqrt{4096} \\ 64 \end{array}$$

$$\begin{array}{r} \sqrt{8836} \\ 94 \end{array}$$

$$\begin{array}{r} \sqrt{12996} \\ 114 \end{array}$$

* Lower - order:

$$\begin{array}{r} \sqrt{3481} \\ \rightarrow 59 \end{array}$$

$$\begin{array}{r} \sqrt{7921} \\ 89 \end{array}$$

$$\begin{array}{r} \sqrt{2304} \\ 48 \end{array}$$

$$\begin{array}{r} \sqrt{7744} \\ 88 \end{array}$$

$$\begin{array}{r} \sqrt{13689} \\ 117 \end{array}$$

$$\begin{array}{r} \sqrt{9409} \\ 97 \end{array}$$

$$\begin{array}{r} \sqrt{18496} \\ 136 \end{array}$$

$$\begin{array}{r} \sqrt{9216} \\ 96 \end{array}$$

$$\begin{array}{r} \sqrt{4356} \\ 66 \end{array}$$

* Upper - Lower ?????

$$\begin{array}{r} \sqrt{2601} \\ \rightarrow \begin{array}{c} \textcircled{25} \\ 36 \end{array} \rightarrow 59 \end{array}$$

$$\begin{array}{r} \sqrt{14161} \\ \begin{array}{c} 121 \\ \rightarrow 119 \\ \textcircled{144} \end{array} \end{array}$$

$$\begin{array}{r} \sqrt{4624} \\ 36 \\ \rightarrow 68 \\ 49 \end{array}$$

$$\begin{array}{r} \sqrt{10816} \\ 100 \\ \rightarrow 104 \\ 121 \end{array}$$

* Find square roots of given nos: ①

$$\sqrt{28224}$$

168

$$\sqrt{7921}$$

89

$$\sqrt{23409}$$

153

$$\sqrt{15129}$$

123

$$\sqrt{1764}$$

42

$$\sqrt{21316}$$

146

$$\sqrt{21904}$$

148

$$\sqrt{19044}$$

138

$$\sqrt{1936}$$

~~136~~ 44

$$\sqrt{7569}$$

87

$$\sqrt{26896}$$

164

$$\sqrt{20449}$$

143

$$\sqrt{2209}$$

47

$$\sqrt{9025}$$

95

$$\sqrt{26244}$$

162

$$\sqrt{9801}$$

99

$$\sqrt{2801}$$

51

$$\sqrt{13924}$$

118

②:

$$\sqrt{1681}$$

41

$$\sqrt{17956}$$

134

$$\begin{array}{r} \sqrt{6889} \\ 83 \end{array}$$

$$\begin{array}{r} \sqrt{2304} \\ 48 \end{array}$$

$$\begin{array}{r} \sqrt{28561} \\ 169 \end{array}$$

$$\begin{array}{r} \sqrt{7225} \\ 85 \end{array}$$

$$\begin{array}{r} \sqrt{25281} \\ 159 \end{array}$$

$$\begin{array}{r} \sqrt{8464} \\ 92 \end{array}$$

$$\begin{array}{r} \sqrt{7744} \\ 88 \end{array}$$

$$\begin{array}{r} \sqrt{19341} * \\ 139 \end{array}$$

$$\begin{array}{r} \sqrt{12769} \\ 113 \end{array}$$

$$\begin{array}{r} \sqrt{19881} \\ 141 \end{array}$$

$$\begin{array}{r} \sqrt{2116} \\ 46 \end{array}$$

$$\begin{array}{r} \sqrt{1521} \\ 39 \end{array}$$

$$\begin{array}{r} \sqrt{9409} \\ 97 \end{array}$$

$$\begin{array}{r} \sqrt{14641} \\ 121 \end{array}$$

$$\begin{array}{r} \sqrt{24336} * \\ 156 \end{array}$$

$$\begin{array}{r} \sqrt{1369} \\ 37 \end{array}$$

③ : from squares of nos!

$$\begin{array}{r} \sqrt{1296} \\ 36 \end{array}$$

$$\begin{array}{r} \sqrt{208224} \\ 456 \end{array}$$

$$\begin{array}{r} \sqrt{2401} \\ 49 \end{array}$$

$$\begin{array}{r} \sqrt{10404} \\ 102 \end{array}$$

$$\begin{array}{r} \sqrt{5625} \\ 75 \end{array}$$

$$\begin{array}{r} \sqrt{7569} \\ 87 \end{array}$$

$$\begin{array}{r} \sqrt{24649} \\ 157 \end{array}$$

$$\begin{array}{r} \sqrt{7396} \\ 86 \end{array}$$

$$\begin{array}{r} \sqrt{6561} \\ 81 \end{array}$$

$$\begin{array}{r} 256 \\ -243 \\ \hline 13 \end{array}$$

$$\begin{array}{r} 225 \\ -210 \\ \hline 15 \end{array}$$

12

$$\begin{array}{r} 345 \\ -324 \\ \hline 21 \end{array}$$

$$\begin{array}{r} 361 \\ -324 \\ \hline 37 \end{array}$$

$$\begin{array}{r} \sqrt{8649} \\ 93 \end{array}$$

$$\begin{array}{r} \sqrt{5929} \\ 77 \end{array}$$

$$\begin{array}{r} \sqrt{6400} \\ 80 \end{array}$$

$$\begin{array}{r} \sqrt{4356} \\ 66 \end{array}$$

$$\begin{array}{r} \sqrt{15376} \\ 124 \end{array}$$

$$\begin{array}{r} \sqrt{14841} \\ 121 \end{array}$$

$$\begin{array}{r} \sqrt{1444} \\ 38 \end{array}$$

$$\begin{array}{r} \sqrt{10201} \\ 101 \end{array}$$

$$\begin{array}{r} \sqrt{6084} \\ 78 \end{array}$$

$$\begin{array}{r} \sqrt{9409} \\ 97 \end{array}$$

$$\begin{array}{r} \sqrt{19321} \\ 139 \end{array}$$

$$\begin{array}{r} \sqrt{30625} \\ 175 \end{array}$$

$$\begin{array}{r} \sqrt{12769} \\ 113 \end{array}$$

$$\begin{array}{r} \sqrt{20164} \\ 142 \end{array}$$

$$\begin{array}{r} \sqrt{31329} \\ 177 \end{array}$$

$$\begin{array}{r} \sqrt{141691} \\ 119 \end{array}$$

$$\begin{array}{r} \sqrt{27889} \\ 167 \end{array}$$

$$\begin{array}{r} \sqrt{34596} \\ 184 \end{array}$$

$$\begin{array}{r} \sqrt{19044} \\ 138 \end{array}$$

$$\begin{array}{r} \sqrt{12544} \\ 112 \end{array}$$

$$\begin{array}{r} \sqrt{29929} \\ 173 \end{array}$$

$$\begin{array}{r} \sqrt{5929} \\ 77 \end{array}$$

$$\begin{array}{r} \sqrt{15625} \\ 125 \end{array}$$

$$\begin{array}{r} \sqrt{25921} \\ 159 \end{array}$$

$$\begin{array}{r} \sqrt{4225} \\ 65 \end{array}$$

$$\begin{array}{r} \sqrt{10609} \\ 103 \end{array}$$

$$\begin{array}{r} \sqrt{33856} \\ 184 \end{array}$$

$$\begin{array}{r} \sqrt{13689} \\ 117 \end{array}$$

$$\begin{array}{r} \sqrt{20436} \\ 156 \end{array}$$

$$\begin{array}{r} \sqrt{35721} \\ 189 \end{array}$$

$$\begin{array}{r} \sqrt{18225} \\ 135 \end{array}$$

$$\begin{array}{r} \sqrt{22801} \\ 151 \end{array}$$

$$\begin{array}{r} \sqrt{14161} \\ 119 \end{array}$$

$$\begin{array}{r} \sqrt{20164} \\ 142 \end{array}$$

$$\begin{array}{r} \sqrt{25281} \\ 159 \end{array}$$

$$\begin{array}{r} \sqrt{12321} \\ 111 \end{array}$$

#4 Find the squares from square roots 1

$$\begin{array}{r} 63 \quad 3 \checkmark \\ 3969 \end{array}$$

$$\begin{array}{r} 121 \quad 2 \checkmark \\ 14641 \end{array}$$

$$\begin{array}{r} 132 \quad 5 \checkmark \\ 17424 \end{array}$$

$$\begin{array}{r} 84 \quad 1 \checkmark \\ 7056 \end{array}$$

$$\begin{array}{r} 71 \quad 1 \checkmark \\ 5041 \end{array}$$

$$\begin{array}{r} 101 \quad 2 \checkmark \\ 10201 \end{array}$$

$$\begin{array}{r} 42 \quad 1 \checkmark \\ 1764 \end{array}$$

$$\begin{array}{r} 92 \quad 3 \checkmark \\ 8464 \end{array}$$

$$\begin{array}{r} 53 \quad 3 \checkmark \\ 2809 \end{array}$$

$$\begin{array}{r} 133 \quad 7 \checkmark \\ 17689 \end{array}$$

$$\begin{array}{r} 64 \quad 8 \checkmark \\ 4096 \end{array}$$

$$\begin{array}{r} 94 \quad 7 \checkmark \\ 8836 \end{array}$$

$$\begin{array}{r} 114 \quad 8 \checkmark \\ 12996 \end{array}$$

$$\begin{array}{r} 59 \quad 3 \checkmark \\ 3481 \end{array}$$

$$\begin{array}{r} 89 \quad 11 \checkmark \\ 7921 \end{array}$$

$$\begin{array}{r} 46 \quad 3 \checkmark \\ 2116 \end{array}$$

$$\begin{array}{r} 88 \quad 10 \checkmark \\ 7744 \end{array}$$

$$\begin{array}{r} 117 \quad 4 \checkmark \\ 13689 \end{array}$$

$$\begin{array}{r} 97 \quad 13 \checkmark \\ 9409 \end{array}$$

$$\begin{array}{r} 136 \quad 11 \checkmark \\ 18496 \end{array}$$

$$\begin{array}{r} 96 \quad 3 \checkmark \\ 9216 \end{array}$$

$$\begin{array}{r} 66 \quad 3 \checkmark \\ 4356 \end{array}$$

$$\begin{array}{r} 59 \quad 8 \checkmark \\ 3481 \end{array}$$

$$\begin{array}{r} 119 \quad 8 \checkmark \\ 14161 \end{array}$$

$$\begin{array}{r} 68 \quad 10 \checkmark \\ 4624 \end{array}$$

$$\begin{array}{r} 104 \quad 10 \checkmark \\ 10816 \end{array}$$

$$\begin{array}{r} 89 \quad 15 \checkmark \\ 17921 \end{array}$$

$$\begin{array}{r} 153 \quad 9 \checkmark \\ 23409 \end{array}$$

$$\begin{array}{r} 123 \quad 7 \checkmark \\ 15129 \end{array}$$

$$\begin{array}{r} 42 \quad 1 \checkmark \\ 1764 \end{array}$$

$$\begin{array}{r} 146 \quad 17 \checkmark \\ 21316 \end{array}$$

$$\begin{array}{r} 48 \quad 6 \checkmark \\ 2304 \end{array}$$

$$\begin{array}{r} 138 \quad 4 \checkmark \\ 19044 \end{array}$$

$$\begin{array}{r} 117 \\ \times 2 \\ \hline 234 \end{array}$$

$$\begin{array}{r} 144 \\ \times 2 \\ \hline 288 \end{array}$$

$$\begin{array}{r} 44 \\ 1936 \end{array}$$

$$\begin{array}{r} 87 \\ 7569 \end{array}$$

$$\begin{array}{r} 164 \\ 26896 \end{array}$$

$$\begin{array}{r} 143 \\ 20449 \end{array}$$

$$\begin{array}{r} 47 \\ 2209 \end{array}$$

$$\begin{array}{r} 95 \\ 7225 \end{array}$$

$$\begin{array}{r} 152 \\ 26244 \end{array}$$

$$\begin{array}{r} 99 \\ 9801 \end{array}$$

$$\begin{array}{r} 51 \\ 2601 \end{array}$$

$$\begin{array}{r} 118 \\ 13924 \end{array}$$

$$\begin{array}{r} 41 \\ 1681 \end{array}$$

$$\begin{array}{r} 137 \\ 17956 \end{array}$$

$$\begin{array}{r} 83 \\ 6889 \end{array}$$

$$\begin{array}{r} 48 \\ 2304 \end{array}$$

$$\begin{array}{r} 169 \\ 29561 \end{array}$$

$$\begin{array}{r} 85 \\ 7225 \end{array}$$

$$\begin{array}{r} 159 \\ 25281 \end{array}$$

$$\begin{array}{r} 92 \\ 8464 \end{array}$$

$$\begin{array}{r} 88 \\ 7744 \end{array}$$

$$\begin{array}{r} 139 \\ 19321 \end{array}$$

$$\begin{array}{r} 113 \\ 12769 \end{array}$$

$$\begin{array}{r} 141 \\ 19881 \end{array}$$

$$\begin{array}{r} 46 \\ 2116 \end{array}$$

$$\begin{array}{r} 39 \\ 1521 \end{array}$$

$$\begin{array}{r} 97 \\ 9409 \end{array}$$

$$\begin{array}{r} 121 \\ 14641 \end{array}$$

$$\begin{array}{r} 156 \\ 19336 \end{array}$$

$$\begin{array}{r} 37 \\ 1369 \end{array}$$